THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET A

Chapter 11: **Areas Related to Circles** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Use $\pi = \frac{22}{7}$ unless stated otherwise.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The area of a sector of a circle of radius 14 cm and central angle 45° is: 1

(a) 77 cm² **(b)** 154 cm² **(c)** 38.5 cm² **(d)** 308 cm²

2. The length of an arc of a circle of radius 14 cm subtending 30° at the centre is: 1

(a) $\frac{22}{3}$ cm **(b)** $\frac{44}{3}$ cm **(c)** $\frac{11}{3}$ cm **(d)** 22 cm

3. The area of a quadrant of a circle whose circumference is 22 cm is: 1

(a) $\frac{77}{8}$ cm² **(b)** $\frac{77}{2}$ cm² **(c)** $\frac{77}{4}$ cm² **(d)** 154 cm²

4. If the area of a sector is $\frac{1}{12}$ of the area of the circle, its central angle is: 1

(a) 30° **(b)** 12° **(c)** 60° **(d)** 36°

5. A student states: "The area of a sector of angle θ is $\frac{\theta}{360^\circ} \times 2\pi r$." The correct response is: 1

(a) Correct

(b) Incorrect — that is the arc length; area $= \frac{\theta}{360^\circ}\pi r^2$

(c) Correct only for a semicircle

(d) Cannot be determined

6. The angle described by the minute hand of a clock in 5 minutes is: 1

(a) 30° **(b)** 5° **(c)** 60° **(d)** 6°

7. For a circle of radius 7 cm with a minor sector of 60°, Student A says the major sector area is $\frac{300}{360}\pi r^2$; Student B says it is $\frac{60}{360}\pi r^2$. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

8. The perimeter of a semicircular protractor of diameter 14 cm is: 1

(a) 22 cm **(b)** 36 cm **(c)** 44 cm **(d)** 29 cm

9. The arc length of a sector of radius 10 cm is 5π cm. Its area is: 1

(a) 25π cm² **(b)** 50π cm² **(c)** 10π cm² **(d)** 5π cm²

10. The area of the minor segment of a circle of radius r with central angle 90° is: 1

(a) $\frac{\pi r^2}{4} - \frac{r^2}{2}$ **(b)** $\frac{\pi r^2}{4} + \frac{r^2}{2}$ **(c)** $\frac{\pi r^2}{2} - r^2$ **(d)** $\frac{\pi r^2}{4}$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the area of a sector of a circle of radius 4 cm with central angle 30°. (Use $\pi = 3.14$) 2

12. The length of the minute hand of a clock is 14 cm. Find the area swept by it in 5 minutes. 2

13. An umbrella has 8 ribs which are equally spaced. Assuming it to be a flat circle of radius 45 cm, find the area between two consecutive ribs. 2

14. A student claims that an arc subtending 60° in a circle of radius 21 cm is longer than the radius. Verify. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A chord of a circle of radius 10 cm subtends a right angle at the centre. Find the areas of the minor and major segments. (Use $\pi = 3.14$) 3

16. To warn ships of underwater rocks, a lighthouse spreads a red light over a sector of angle 80° to a distance of 16.5 km. Find the area of the sea over which ships are warned. (Use $\pi = 3.14$) 3

17. An arc of a circle of radius 35 cm subtends 72° at the centre. Find 3
(i) the length of the arc
(ii) the area of the sector
(iii) the perimeter of the sector.

OR

A chord of a circle of radius 12 cm subtends an angle of 120° at the centre. Find the area of the corresponding minor segment. (Use $\pi = 3.14$, $\sqrt{3} = 1.73$) 3

18. A pendulum swings through an angle of 30° and describes an arc 8.8 cm long. Find the length of the pendulum. 3

OR

The perimeter of a sector of a circle of radius 5.2 cm is 16.4 cm. Find the area of the sector. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. In a circle of radius 28 cm, a chord AB subtends a right angle at the centre O. Find 5
(i) the length of arc AB
(ii) the area of the minor sector
(iii) the area of the minor segment
(iv) the area of the major segment.

OR

Case study: A wall clock has a minute hand 21 cm long and an hour hand 14 cm long. Find 5
(i) the area swept by the minute hand in 20 minutes
(ii) the distance moved by the tip of the hour hand in 2 hours
(iii) the area swept by the hour hand in 3 hours.

20. **Case study:** A cow is tied with a 14 m rope at one corner of a rectangular field measuring 20 m × 16 m. 5
(i) Find the area of the field the cow can graze.
(ii) Find the increase in grazing area if the rope were 16 m long.
(iii) Find the area of the field the cow cannot graze with the 14 m rope.

OR

Find the area of the minor segment of a circle of radius 14 cm if the length of the corresponding arc is 22 cm. Also find the area of the major segment. 5

END OF QUESTION PAPER Total: 40 Marks